ROBERTO GUAGNELLI GARCÍA

robguagnelli@gmail.com | Querétaro, Mexico
linkedin.com/in/roberto-guagnelli-garcia | robertoguagnelli.github.io

Mechanical Engineering | **Aeronautical Engineering** | **Aircraft Design & Systems Integration**

Spanish and Mexican citizen | EU work authorization | Graduating July 2027 | Available August 2027 | Open to relocation

EDUCATION

Tecnológico de Monterrey, Campus Estado de México — Expected Jul 2027
B.S. Mechanical Engineering; Concentration in Aeronautical Engineering
Academic average: 91/100

WORK EXPERIENCE

GE Aerospace | Designer Intern, Product Definition — Aug 2026 – Present
RISE | Siemens NX / Teamcenter

- Designed **70+ support brackets** and completed **20+ electrical harness routing revisions** for mechanical/electrical integration, considering manufacturing, installation and maintainability.
- Resolved clearances, adjacent-system interferences and minimum cable bend-radius issues; prepared CAD models and product-definition drawings using established datums and tolerance tables.
- Work directly with vibration-analysis engineering teams in **Mexico and the United States** to incorporate mechanical interface and vibration/fatigue requirements into design iterations.

GE Aerospace | Lean Summer Experience Intern — Jun 2026 – Jul 2026
Integrated Performance & Operability (IP&O)

- Developed visual planning tools, KPI dashboards and weekly reviews with program leads; meeting leaders measured **approximately 20% shorter meetings**. Supported process automation in Python.

ENGINEERING PROJECTS & LEADERSHIP

Borregos CEM AeroDesign | Tecnológico de Monterrey — Sep 2023 – Present
Engineering & Leadership Advisor (current); former Design Captain / Engineering Coordinator — Aug 2024 – Present

- Led **five designers** and coordinated aircraft integration across structures, electronics, design and R&D within a **40+ member team**; helped deliver **three aircraft built and flown**.
- Personally modeled **300+ CAD parts per aircraft** within 400+ component assemblies, including aerodynamic geometry, structures, landing gear, mechanical joints and system-installation hardware.
- **Wingull (2025):** Designed removable joints for a **4.2 m wing** to meet a **1.2 m transport limit**; supported a physical wing-load test. The aircraft completed payload-carrying flight.
- **Latías (2026):** Built a longitudinal flight-dynamics model with **Python and XFLR5**, calculating stability derivatives and state-space eigenvalues to characterize phugoid/short-period modes; revised payload placement to recover modeled dynamic stability.
- Performed aircraft-level aerodynamic, stability and propulsion trade studies; used CFD and XFLR5 findings to inform aircraft design and takeoff-performance modeling.

Manufacturing Coordinator | Hyperion — Sep 2023 – Aug 2024

- Coordinated **eight builders** in fabrication, assembly and quality control; designed and manufactured a spring-loaded aluminum nose landing gear, fitted carbon-fiber fuselage skin and corrected rib alignment with an assembly spacer.
- Supported an aircraft that completed **six competition flights carrying 3 kg payload**; team placed **2nd overall** and **3rd in design report** at SAE Aero Design México 2024.

SELECTED ENGINEERING WORK

- **Wing CFD (2026):** Developed a 3D ANSYS Fluent workflow covering mesh refinement, SST k–ω, steady/transient simulation, flow visualization and comparison with XFLR5 to guide wing-incidence and tip-design decisions.
- **Extruder failure investigation (2026):** Evaluated a fractured tool-steel screw using hardness testing, metallography, SEM/EDS evidence and torsional stress calculations.

TECHNICAL SKILLS

CAD & Analysis: Siemens NX, Teamcenter, SOLIDWORKS, ANSYS Fluent, XFLR5; engineering drawings, 3D CAD, steady/transient CFD, flight dynamics.
Programming & Data: Python; engineering analysis and process automation.
Engineering & Manufacturing: Aircraft configuration, systems integration, mechanical/electrical interfaces, structural test support, composites, 3D printing, laser cutting, aluminum welding.
Certification: Certified SOLIDWORKS Professional (CSWP).
Languages: Spanish (native), English (C2), German (A2).